



CS 350: Computer Graphics

Course Code	CS 350
Course Name	Computer Graphics
L-T-P-C	3-0-0-3
Approval	5th Senate (Oct 2013)

Course Contents

Raster Graphics; line and circle drawing algorithms; Windowing and 2D/3D clipping. Cohen and Sutherland line clipping, Cyrus Beck clipping method; 2D and 3D Geometrical Transformations: scaling, translation, rotation, reflection; Viewing Transformations: parallel and perspective projection; Curves and Surfaces: cubic splines, Bezier curves, B-splines, Parametric surfaces. Surface of revolution Sweep surfaces, Fractal curves and surfaces; Hidden line/surface removal methods; illuminations model; shading, Introduction to Ray-tracing; Animation; Programming practices with standard graphics libraries like OpenGL.

Text and Reference Books

1. Edward Angel, **Interactive Computer Graphics: A top-down approach using OpenGL**, 5th Edition, Pearson Education.
2. Hearn and Baker, **Computer Graphics using OpenGL**, 3rd Editoin, Pearson Education.
3. Foley, Dam, Feiner and Hughes, **Computer Graphics: Principles and Practice in C**, Pearson Education.
4. Hill, **Computer Graphics using OpenGL**, PHI.